Yusuke Ozaki

Kwansei Gakuin University · RIKEN Center for Biosystems Dynamics Research (BDR) / RIKEN AGIS
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Research interests: Bayesian optimization, bandits, and active learning for self-driving laboratories

Education

Kwansei Gakuin University Apr 2022 – Mar 2027 (expected)
B.Eng. in Computer Science, School of Engineering Sanda, Hyogo, Japan

- GPA: 3.62/4.00
- Research advisor: Akihiro Inokuchi (Professor)

University at Albany, State University of New York Aug 2025 – May 2026
Exchange Student, Department of Computer Science Albany, NY, USA

- GPA: 4.00/4.00
- Research advisor: Chong Liu (Assistant Professor)

Publications & Manuscripts

Workshop papers

[W1] **Bayesian Optimization of Protocols**
Yusuke Ozaki, Kazunari Kaizu, and Koichi Takahashi
NeurIPS 2026 Workshop on AI for Drug Discovery (AI4DD), 2026.

Under review

[R1] **Object-Flow Machine Learning: Operation-Level Surrogate Modeling for Self-Driving Laboratories**
Yusuke Ozaki, Kazunari Kaizu, and Koichi Takahashi
Under review.

[R2] **Cumulative Regret Analysis of the Piyavskii–Shubert Algorithm in Lipschitz Bandits**
Yusuke Ozaki and Chong Liu
Under review.

[R3] **SPIN: Structural LLM Planning via Iterative Navigation for Industrial Tasks**
Yusuke Ozaki and Dhaval Patel
Under review.
Preprint: arXiv:2605.14051, 2026.

Research Experience

RIKEN Center for Biosystems Dynamics Research (BDR) Jun 2024 – Aug 2025, Jun 2026 – Present
Research Part-timer II, Lab. for Biologically Inspired Computing (PI: Koichi Takahashi) Kobe, Japan
Also affiliated with the RIKEN AGIS program (since Jun 2026).

- **Object-Flow Machine Learning (OFML).** Proposed an operation-level surrogate model that represents each experimental protocol as a typed DAG with one shared module per operation type, enabling zero-shot prediction and transfer across protocols. Proved uniform convergence of the posterior mean under noisy variance-reduction (IVR) active learning and derived zero-shot conditions. Under review [R1].
- **ProtocolUCB.** Proposed a UCB method over protocol–condition pairs that composes shared Bayesian operation models; proved that its recommendations approach the set of global maximizers almost surely. Across 16 protocols, sharing reduced mean protocol-wise simple regret by 22% at $T = 40$. Accepted at the NeurIPS 2026 AI4DD workshop.
- **Lab automation.** Develop SiLA 2 connectors for a Tecan Fluent 780 (with Inheco ODTC, Infinite 200 PRO, BioShake), a Yaskawa Maholo LabDroid, and a Franka Research 3 arm. Ran closed-loop active learning with OFML on the physical Fluent (11 iterations, Aug 2026); currently automating *E. coli* transformation on it. In 2024–2025, built ML-based experiment simulators for automated liquid handling.

University at Albany, SUNY, Department of Computer Science (Advisor: Chong Liu) Aug 2025 – Present
Undergraduate Researcher Albany, NY, USA

- **Lipschitz bandits.** Proved that the cumulative regret of the Piyavskii–Shubert algorithm matches a lower bound in order for every dimension in the noiseless setting. Proposed LiP-UCB for the noisy setting, with a high-probability regret bound whose expectation matches a Gaussian-noise lower bound up to a log factor. Under review [R2].

- **Preference-based BO** (Aug 2025 – Jan 2026). Studied Bayesian optimization with a Generalized Preference Model, analyzing regret bounds and identifiability when learned embeddings are combined with Gaussian-process models.
- **Quantum algorithms for BO** (Jun 2026 – present; with Chaowen Guan, University of Cincinnati). Investigate quantum algorithms to accelerate the linear algebra in Gaussian-process Bayesian optimization.

Kwansei Gakuin University, Inokuchi Lab (Advisor: Akihiro Inokuchi) Aug 2024 – Present
Undergraduate Researcher Sanda, Hyogo, Japan

- Designed a graph neural network that learns task-specific sparse adjacency layer by layer, with an actor–critic edge-selection module trained by policy gradient, to reduce the edges used in convolution.
- Writing an undergraduate thesis on cumulative regret in Lipschitz bandits (due Feb 2027).

University at Albany, SUNY, with IBM (Collaborator: Dhaval Patel, IBM) Aug 2025 – May 2026
Capstone Team Lead, then Research Collaborator Albany, NY, USA

- Led a senior capstone team whose project grew into a year-long research collaboration with IBM on LLM agents for industrial asset operations.
- Developed SPIN, a structural planning method for LLM agents, and built its evaluation stack (agent prompts, trajectory logging, automatic DAG/plan evaluation). Released as an arXiv preprint (May 2026); under review [R3].

The University of Osaka, PRIMe, Yachie Lab (PI: Nozomu Yachie) Feb 2025 – Aug 2025
Research Assistant Osaka, Japan

- Built pipelines for gene regulatory network inference from single-cell RNA-seq and ATAC-seq data.

Presentations

- How to Realize Bayesian Optimization for Self-Driving Laboratories, YAML 2026 (Young Researchers' Association for Machine Learning), Japan (Oral, in Japanese).
- Object-Flow Machine Learning: Active learning framework utilizing protocols information, AutoML 2025, New York City, USA (Poster).
- Development of a Machine Learning Method Utilizing Experimental Protocols to Predict Outcomes and an Active Learning Framework Employing the Method, RIKEN BDR Student Symposium 2025, Kobe, Japan (Oral).
- Development of a Machine Learning Method Utilizing Protocols to Predict Experimental Outcomes, Spring School for Theoretical Biology 2025, Hiroshima, Japan (Oral).

Software

- **tecan-fluent**: SiLA 2 connector for a Tecan Fluent 780 (arm, gripper, deck cameras, thermal cycler, plate reader, shaker), built with the UniteLabs CDK.
- **inheco-odtc**: SiLA 2 connector for an Inheco ODTC 96 thermal cycler integrated in the Fluent.
- **cx_bayes_al**: Lean 4 formalization and experiments for global consistency of cost-aware Bayesian active learning.
- **MEG**: Minimal Edges Generator, a sparse-adjacency graph neural network.

Honors and Awards

Gyomu Super Japan Dream Foundation Study Abroad Scholarship Aug 2025 – Jul 2026

Keyence Foundation "Ganbare! Nihon no Daigakusei" Student Support Grant Jul 2024